

COMBINATORIAL ANALYSIS: COUNTING THE POSSIBLE

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HOW TO COUNT EFFICIENTLY

In Article No.8, we introduced probability theory as a mathematical framework for quantifying uncertainty. However, before we can calculate probabilities, we must first determine the number of different ways an event can occur. The mathematic theory of counting, arranging, and selecting objects is referred to as combinatorial analysis, or combinatorics. While counting may seem like a simple task, many real world problems involve such a large number of possibilities that direct enumeration is impractical. By developing systematic counting techniques, combinatorial analysis allows us to solve complex problems efficiently.

The importance of combinatorial analysis extends far beyond mathematics. It forms the foundation of many probability calculations and has applications in fields such as computer science, artificial intelligence, cryptography, and quantitative finance. Whether we are counting the number of possible poker hands, analysing trading scenarios, or constructing probabilistic models, combinatorial methods provide the help needed to understand the underlying structure of the problem. In this article, I introduce the fundamental principles of combinatorial analysis, including permutations and combinations, and explore how these concepts can be used to tackle probability problems.

FUNDAMENTAL COMBINATORIAL ANALYSIS

We must first explore the basic principle of counting. If we let S be a set of length r sequences (which is a collection of ordered sequences, where each sequence contains r elements), where there are n_1 possible first entries, n_2 possible second entries for each first entry, n_3 possible third entries for each combination of first and second entries, etc., thus there are a total of $n_1 \cdot n_2 \cdot \dots \cdot n_r$ possible outcomes.

A **permutation** is a rearrangement of objects into a distinct sequence, where order matters. There are $\frac{n!}{n_1!n_2!\dots n_r!}$ different permutations of n objects, of which n_1 are alike, n_2 are alike, ..., n_r are alike. A **combination** is an unordered collection of objects, where order doesn't matter. There are $\binom{n}{r} = \frac{n!}{(n-r)!r!}$ different combinations of n distinct objects taken r at a time.

Combinations appear naturally when expanding algebraic expressions. Consider the expansion of $(a+b)^n$. Each term is formed by choosing either a or b from each of the n factors. The coefficient of $a^r b^{n-r}$ is therefore the number of ways to choose r of the n factors to contribute an a , namely $\binom{n}{r}$, referred to as n choose r .

This leads to the **Binomial theorem**: $(a+b)^n = \sum_{r=0}^n \binom{n}{r} a^r b^{n-r}$. The theorem provides a connection between combinatorial counting and algebra, which arises frequently in probability theory, particularly when analysing binomial distributions.

In many counting problems, simply adding the sizes of different sets can lead to over counting when the sets overlap. The **Inclusion-Exclusion Principle** provides a way to correct for this. For two sets, A_1 and A_2 , $P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2)$. The intersection is subtracted because elements belonging to both sets are counted twice in the sum. More generally, for several overlapping sets, inclusion-exclusion alternates between adding and subtracting intersections of increasing size. Therefore, $P(A_1 \cup A_2 \cup A_3) = P(A_1) + P(A_2) + P(A_3) - P(A_1 \cap A_2) - P(A_1 \cap A_3) - P(A_2 \cap A_3) + P(A_1 \cap A_2 \cap A_3)$, and even more generally, $P(A_1 \cup A_2 \cup \dots \cup A_n) = \sum_{i=1}^n P(A_i) - \sum_{i_1 < i_2} P(A_{i_1} \cap A_{i_2}) + \dots + (-1)^{r+1} \sum_{i_1 < i_2 < \dots < i_r} P(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_r}) + \dots + (-1)^{n+1} P(A_1 \cap A_2 \cap \dots \cap A_n)$, where $\sum_{i_1 < i_2 < \dots < i_r} P(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_r})$ has $\binom{n}{r}$ terms. This principle is an essential tool in combinatorial analysis and probability, where we are often interested in counting outcomes that satisfy at least one of several conditions.

AN EXAMPLE: THE POKER HAND PROBLEM

A classic example in combinatorial analysis is the poker hand problem, which provides an excellent illustration of many fundamental counting techniques. Poker is a card game in which each player receives a hand of 5 cards. There are 52 cards in a deck. Each card has a value and belongs in a suit. There are 13 values, 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K, A, and 4 suits ♠, ♣, ♥, ♦. What are the probabilities of getting hands with a four-of-a-kind (4 of the 5 cards with the same value)? Hands with a full house (3 cards of 1 value and 2 cards of another value)? Hands with 2 pairs?

Solution: First, let us determine the total number of hands which is possible. The number of different hands of a 5 card draw is the number of 5 element subsets of a 52 element set, thus is $\binom{52}{5} = 2,598,960$.

Hands with a four-of-a-kind: To choose the value of the 4 cards with the same value, there are 13 choices. The 5th card can be any of the remaining 48 cards (12 choices for values and 4 choices for suits). Thus, the number of hands with a four-of-a-kind is $13 \times 48 = 624$.

Hands with a full house: In sequence, we need to choose the value of the triple, 13 choices; the suits of the triple, $\binom{4}{3}$ choices; the value of the pair, 12 choices; and the suits of the pair, $\binom{4}{2}$ choices. Thus, the number of hands with full house is $13 \times \binom{4}{3} \times 12 \times \binom{4}{2} = 13 \times 4 \times 12 \times 6 = 3,744$.

Hands with 2 pairs: In sequence, we need to choose the values of the 2 pairs, $\binom{13}{2}$ choices; the suits of the first pair, $\binom{4}{2}$ choices; the suits of the second pair, $\binom{4}{2}$ choices; and the remaining card, 44 ($52 - 4 \times 2$, since the last cards cannot have the same value as either pair) choices. Thus, the number of hands with 2 pairs is $\binom{13}{2} \times \binom{4}{2} \times \binom{4}{2} \times 44 = 78 \times 6 \times 6 \times 44 = 123,552$.

To calculate the probability of each, we must divide the number of hands of each kind by the total possible number of hands. Thus, we obtain $\frac{1}{4165}$, $\frac{6}{4165}$, and $\frac{198}{4165}$ respectively.

WHY COMBINATORIAL ANALYSIS MATTERS IN FINANCE

Combinatorial analysis plays an important role in quantitative finance, as many problems involve counting a large number of possible outcomes. Before we can assign probabilities to different market scenarios, we first need to understand how many scenarios are possible and the ways in which they can occur.

One common application is in option pricing and risk management, where we often consider multiple paths that an asset price may take over time. In discrete time models, such as the binomial option pricing model, combinatorial methods can be used to count these possible price paths and determine their probabilities. Classical combinatorics is primarily concerned with finite or countably infinite sets, where we can literally count outcomes. Combinatorics provides the counting tools for discrete probability models, while continuous finance models replace counting with integration. The latter can often be viewed as the limiting case of the former. Similar concepts are used in portfolio analysis, where investors must assess different combinations of assets and allocation choices.

More broadly, combinatorial analysis appears in areas such as algorithmic trading, portfolio optimisation, and scenario analysis. Many financial problems can ultimately be viewed as questions of counting and arranging outcomes. As a result, combinatorial analysis serves as a bridge between probability theory and its practical application in finance, providing tools that are used throughout quantitative modelling.

CONCLUSION

Combinatorial analysis provides a collection of tools for counting outcomes efficiently. By understanding counting principles, permutations, and combinations, we can tackle complex probability problems that would otherwise be impossible to solve directly. These techniques form an important foundation for more advanced topics in probability and statistics.

ALEXANDER FLANN